Seminar

Mutation Testing and Test Effectiveness





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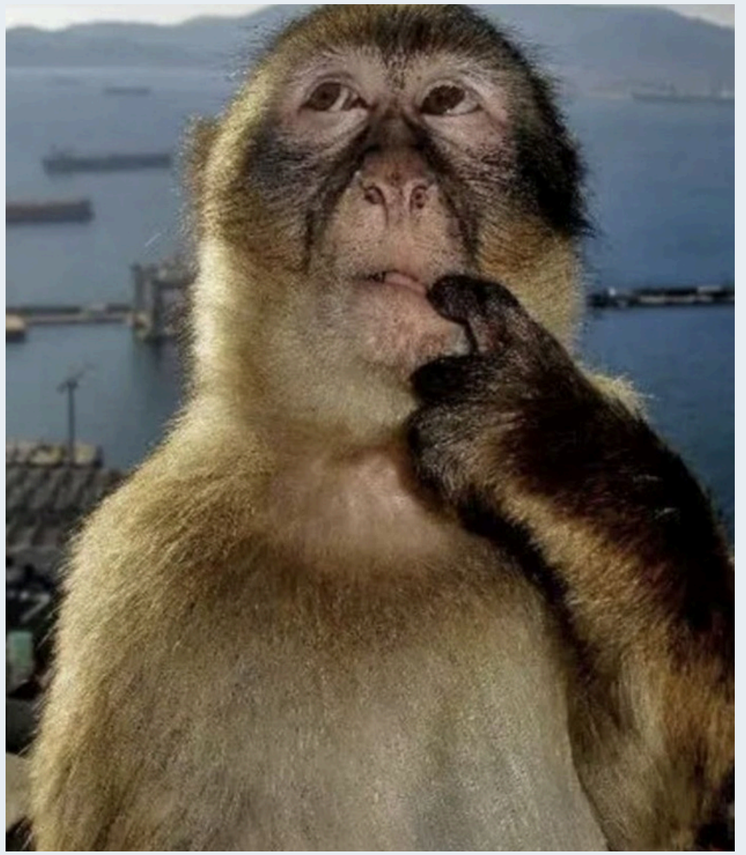
All tests passed. So... no bugs?

ALL TESTS

PASS

Does PASS mean the code is correct?

Question: Can tests catch wrong results?



Running code ≠ checking the result.

CODE COVERAGE

Did we visit this code?

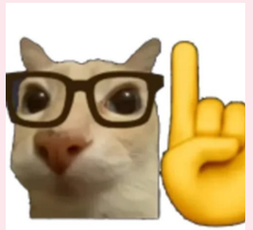
✓ YES



MUTATION TESTING

Did we notice a wrong result?

? NOT ALWAYS



A test can run the right code and still miss the wrong answer.

One character can change the answer.

ORIGINAL RULE

amount ≥ 100

CHANGED RULE

amount > 100

Test the exact boundary: **100**

✓ YES

× NO

The mutant is visible only when the input exposes its difference.

Stryker makes mistakes. Tests hunt them.

01
START
Good code

→

02
MUTATE
Change one thing

→

03
RETEST
Run same tests

→

04
DECIDE
Read result

✓ **KILLED**
A test fails on the changed code

× **SURVIVED**
The tests still pass

A SURVIVOR is a clue to investigate, not proof of a bug.

We tested an online shop backend.

88

PASSING TESTS

Our starting point

36

Cart & Orders

Add items and place orders

26

Coupons

Discount and usage limits

26

Accounts

Login and access

First we wrote passing tests. Then Stryker tested those tests.

Two helpers. Different jobs. Humans decide.

STRYKERJS

Find weak tests

- Create mutants
- Run tests
- Show survivors

AI ASSISTANT

Suggest improvements

- Propose assertions
- Explain candidates
- May be wrong

HUMAN REVIEW

Check what is right

- Read requirements
- Accept or reject
- Verify by running

Why StrykerJS? Our EShop uses JavaScript. PIT targets Java; mutmut targets Python.

The new cart was not really empty.

WHAT THE MUTANT DID

NEW CART

1 unexpected item

It should contain ZERO items.

WHAT THE TEST CHECKED

Before	**“Request succeeded”**
After	**“Cart has 0 items”**

SURVIVED → KILLED

Key idea: “Success” does not mean the DATA is correct.

Better input reveals the hidden difference.

SET USAGE LIMIT = 1

Expected saved value

1 vs 1

NO DIFFERENCE

SET USAGE LIMIT = 3

Expected saved value

3 vs 1

MUTANT DETECTED

Changing the input helped our tests detect 2 related mutants.

Same starting point. Different test improvements.

SAME ORIGINAL CODE + BASELINE TESTS

MANUAL

People add better checks

AI-ASSISTED

AI suggests → human verify

RUN TESTS AGAIN → COMPARE WHICH MUTANTS WERE KILLED

More useful checks caught more changes.

CART & ORDERS



Number of mutants detected

COUPONS

Start	39
Manual	41
AI-assisted	41

Auth: comparison pending

A bigger score is not the whole truth.

01 TIMEOUT

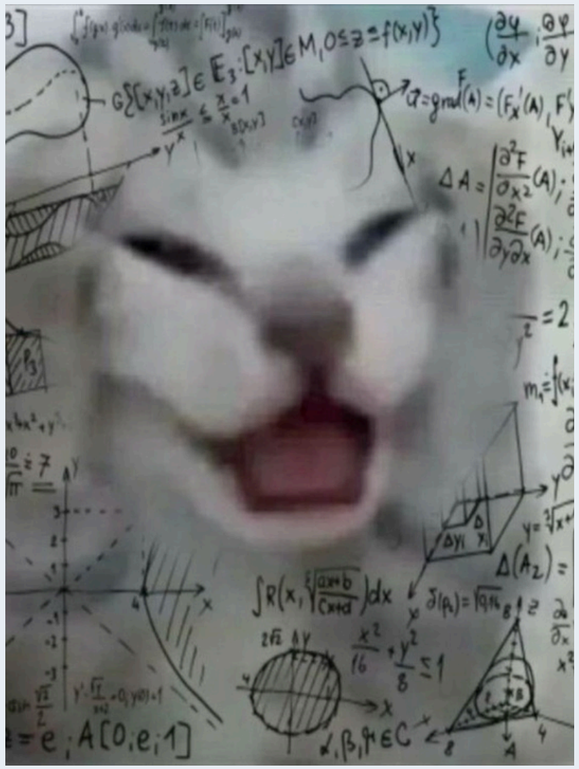
A slow test is not a proven bug catch.

02 WRONG RULE

Tests can enforce bad expectations.

03 SURVIVOR

It might be a missing test or equivalent.



AI gives ideas. A passing test on the original + a verified mutant kill gives evidence.

Now watch us hunt a mutant.

SEE IT CHANGE.

WATCH THE TEST CATCH IT.

NEXT 10 MIN: REAL TERMINAL + ESHOP

Afterward: your turn with 5 mutants.



THE END OF MY PRESENTATION



ANY QUESTION BRUH?